ChancaDEM: A Crusher-Comminution Studio and a Held-Out Evaluation of Cone-Crusher Models on Ten Industrial Surveys

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_ChancaDEM

***Abstract*. A cone crusher turns a feed into a product whose size, throughput and power are set by four levers: the closed-side setting, the eccentric throw, the eccentric speed and the feed size distribution. ChancaDEM is an open studio that couples a Whiten population balance, an Evertsson capacity model and Bond's power law in one engine, runs it in the browser, and holds it against ten measured surveys of a Metso HP500 secondary cone crusher at an itabirite iron-ore operation (Rocha et al., 2024, CC BY). This report states the problem, the data, every method and the evaluation protocol, and reports one benchmark on the same ten surveys: the calibrated engine, a bounded residual correction, a free-form fit, a setting-only line, Morrell's size-specific energy and CrusherBayes, a Bayesian calibration of the engine with a Kennedy and O'Hagan structural discrepancy sampled by differential-evolution moves. Every method is refit without the held-out survey (strict leave-one-survey-out), with a label-shuffle control and a nested 80% predictive band. The setting-only line, with one feature, reaches a held-out mean absolute percentage error of 13.67% on throughput and 12.78% on power; CrusherBayes reaches 12.00% and 14.31%, with 80% bands of coverage 0.80 and 0.90 that are 7% and 44% narrower than the band of the calibrated engine. CrusherBayes does not beat the line on power. The ten surveys identify the capacity scale, the noise and the discrepancy slopes on the setting, and leave the classification, breakage and t_{10} scalars at their prior. Morrell's energy, with its crushing work index and no-load power fitted per fold because the ore's index is not published, reaches 38.08% and, with a no-load term whose fitted index is negative in every fold, 15.21%. The discrete-element package that the product plans is not part of the results.**

***Index Terms*. Comminution, cone crusher, population balance, Bayesian calibration, structural discrepancy, leave-one-out validation, Morrell, Bond, mineral processing.**

I. Problem and Executive Context

CRUSHING is the first stage of comminution, and its setting decides how much work the grinding mill downstream must do. In Chilean copper mining the concentrator took 56,970 TJ, 55.7% of the 102,307 TJ of electricity the sector used in 2024, and the electricity per tonne of fine copper in concentrates rose from 8,946 to 13,079 MJ between 2010 and 2024, an increase of 46.2%, while the national average ore grade fell from 0.75% to 0.62% [1]. The same agency attributes the rise to deeper deposits and harder rock that "directly affect comminution" and projects the electricity of the copper sector to grow by 20.2% from 2025 to 2034, from 27.6 to 33.2 TWh, against 8.3% for fine copper [2]. The decision that follows belongs to the process engineer and the plant operator: at which setting, speed and feed condition to run a crusher, and how much of the size reduction to leave to the mill.

Three outcomes are measured on a plant: the product size (P_{80}, the size passing 80% of the mass), the throughput and the motor power. They depend on the four levers through a non-linear chain: a tighter setting gives a finer product and less capacity, the capacity rises with eccentric speed up to an optimum and falls past it, and a wider throw coarsens the product at a fixed setting. Mechanistic models of this chain exist and are calibrated on survey data of the machine; the practical question is how well a calibrated model predicts a survey it was not fitted to, because the calibrations in the literature are judged on the surveys they were fitted on, and ten surveys leave little room for a free-form correction. The report measures that, with a one-feature baseline and an uncertainty band that a decision maker can read.

II. Data

A. The ten HP500 surveys

The measured data are Table A1 of Rocha et al. [3]: ten surveys of a Metso HP500 secondary cone crusher (CC BY 4.0), each with the closed-side setting (CSS), the feed F_{80}, the feed rate and the motor power. Across the ten surveys the CSS spans 38 to 55 mm (mean 46.9), the feed F_{80} 47 to 140 mm (mean 87.3), the feed rate 813 to 1639 t/h (mean 1145.9) and the power 187 to 355 kW (mean 288.0). The power is correlated with the setting ($r = -0.82$) and with the feed rate ($r = -0.85$), and the feed rate with the setting ($r = 0.81$): a tighter setting draws more power at a lower feed rate. The product size distributions of the paper are figures, not tables, so P_{80} is not measured here and is never a target; the engine's P_{80} at a survey is a model output. The throw (34 mm) and speed (360 rpm) are the nominal values of the machine.

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Interactive studio and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ChancaDEM; the studio is available at https://chancadem.fasl-work.com.

B. The cases

The studio carries 27 cases: 17 circuit cases that span the machine classes (two gyratory, two jaw, six secondary cone, two tertiary cone, two short-head cone, and three controls: a pass-through case, an invalid case and a synthetic anchor) and the ten surveys. The circuit cases are illustrative of their machine class and are not calibrated; only the HP500 lane is calibrated, to one machine on one ore. Every prediction passes a unit check in the bake: it must be finite and lie inside the observable's range widened by a full range on each side, so that a throughput is never read as a power.

III. Related Work and Methods

A. Population balance, capacity and power

The product gradation follows the Whiten classification and breakage balance [4]. For a feed vector $\mathbf{f}$, a classification matrix $\mathbf{C}$ and a breakage matrix $\mathbf{B}$,

$$\mathbf{p} = (\mathbf{I} - \mathbf{C})\,(\mathbf{I} - \mathbf{B}\mathbf{C})^{-1}\,\mathbf{f}. \tag{1}$$

The classification uses the Andersen and Napier-Munn form with the constants K_1, K_2 and K_3, where K_2 tracks the open-side setting [5]. The breakage matrix comes from the JKMRC t_{10} energy-fineness relation and an appearance function [5]. The capacity follows the flow model of Evertsson [6], [7], unimodal in eccentric speed. The power follows Bond's third theory [8], $W = 10\,W_i\,(1/\sqrt{P_{80}} - 1/\sqrt{F_{80}})$ in kWh/t, multiplied by the throughput. The feed is a Rosin–Rammler distribution on a $\sqrt{2}$ sieve grid. The engine exists as a canonical Python implementation and a TypeScript port that runs in the browser; the two agree to a relative difference of 2.3×10^{-12} on a shared reference of 121 operating points, and one evaluation takes about 1.3 ms.

B. The calibration of Rocha et al. as replicated

Rocha et al. [3] fit an Andersen–Whiten model to the ten surveys: $K_1 = 0.23\,\mathrm{CSS} + 0.30\,f_{80}$, $K_2 = 12.0 + 0.55\,\mathrm{CSS} + 0.40\,f_{80}$, $K_3 = 2.3$, $t_{10} = 64 - 0.12\,\mathrm{CSS} - 0.23\,f_{80}$, a capacity base $Q_T \approx 647$ t/h and a power form $P_c = \zeta P_d + P_n$ with $\zeta = 1.30$ and $P_n = 110$ kW. The engine carries these constants (the calibrated backbone) and is run at each survey's measured setting and f_{80}. The paper's power constants apply to its size-specific net power P_d, not to the Bond draw of the engine; applied to the Bond draw they give a mean absolute percentage error of 47.58% in sample, and the bare Bond draw 26.88% (Table I). The Bond draw is nearly flat (196 to 237 kW) against a measured spread of 187 to 355 kW. A least-squares refit of the paper's form on the Bond draw reaches 19.28% in sample with a negative slope $\zeta = -2.41$ and $P_n = 829$ kW, which is a fit to the survey mean and not a physical constant.

C. Setting-only line

The baseline for every method is an ordinary-least-squares line of one target on the closed-side setting alone, $y = a + b\,\mathrm{CSS}$, for throughput and for power, refit on the training surveys of each fold. It carries no physics. Its value is that the claim of any other method is the margin over it.

TABLE I
In-sample power of the calibrated engine against the ten measured values (kW). The paper's constants $\zeta = 1.30$, $P_n = 110$ apply to its size-specific P_d, not to the Bond draw.

Power model	MAPE (%)	RMSE (kW)
Constant mean	22.03	61.65
Bare Bond draw	26.88	93.22
Paper form $1.30\,P_d + 110$ on Bond	47.58	132.85
Refit of the paper form on Bond	19.28	54.97

D. Bounded residual and free-form fits

Two corrections to the calibrated backbone are held out beside it. The bounded residual adds a ridge regression ($\lambda = 1$) on the standardized setting and feed f_{80}, clipped to $\pm 20\%$ of the backbone value. The free-form fit is an unconstrained least-squares regression of the raw target on the same two features. A constant-mean predictor and a label-shuffle control (the residual targets permuted) bound the skill from below.

E. Morrell's size-specific energy

Morrell's method gives the specific energy of a conventional crusher from the feed and product sizes and the crushing work index M_{ic} [9], [10]. With the equations B4 and B5 of the guideline [10],

$$W_c = S_c\,K_2\,M_{ic}\,4\left(x_2^{f(x_2)} - x_1^{f(x_1)}\right)\ [\mathrm{kWh/t}], \tag{2}$$

$$f(x) = -\left(0.295 + x/10^6\right), \qquad S_c = K_s\,(x_1 x_2)^{-0.2}, \tag{3}$$

with x_1 and x_2 the feed and product P_{80} in micrometres, $K_s = 55$, $K_2 = 1.0$ for a crusher in closed circuit with a classifying screen, and S_c applied when it is below one and taken as one above. The power is W_c times the throughput. The index M_{ic} of the Minas Rio itabirite is not published, so it is a fitted scalar, and so is the no-load power; both are refit on the nine training surveys of every fold. The design column of a survey uses the measured f_{80}, the measured feed rate and the engine's P_{80} (a model output used as an input). Two variants are held out: $\mathrm{kW} = M_{ic}\,g$, and $\mathrm{kW} = M_{ic}\,g + P_{noload}$. The primary paper was not opened; the equations are taken from the guideline.

F. Learned surrogate and denoising autoencoder

A multilayer perceptron (11-64-64-32-10) trained on a Latin-hypercube sweep of the engine (6430 training points) emulates its ten outputs, so that what-if exploration is instant; the approach follows the surrogate of a phenomenological comminution model of Koh et al. [11]. A denoising autoencoder (14-16-6-16-14) scores how far a point lies from the training manifold, as a warning that the surrogate is asked to extrapolate. Both are scored on an independent second draw (2196 points, seed 67890), through onnxruntime on the graph the browser runs. Both emulate the engine, not the plant, and neither is a predictor for the HP500 surveys.

TABLE II
PRIORS OF THE ENGINE SCALARS: TRUNCATED NORMAL AROUND THE ENGINE'S OWN VALUES, IN THE LOG SPACE WHERE MARKED. NOMINAL VALUES ARE THOSE OF ROCHA ET AL. [3] (TABLES 1, 4 AND 5); THE WIDTHS ARE THE AUTHOR'S.

Scalar	Unit	Nominal	Prior width
Q_T (capacity base)	t/h	647	log sd 0.25
W_i (blend work index)	kWh/t	9.0	log sd 0.17
ζ	–	1.30	log sd 0.23
P_n (no-load power)	kW	110	sd 50
K_1 multiplier	–	1.0	log sd 0.15
K_2 multiplier	–	1.0	log sd 0.15
K_3	–	2.3	sd 0.5
t_{10} shift	points	0	sd 4

IV. PROPOSED METHOD: CRUSHERBAYES

Rocha et al.'s calibration is a point fit. CrusherBayes places a prior on the engine's fitted scalars, adds an explicit structural discrepancy per target in the form of Kennedy and O'Hagan [12], and returns a posterior predictive distribution for a survey that was not used in the fit. For survey i and target t (throughput in t/h, power in kW),

$$y_{t,i} = f_t(\boldsymbol{\theta};\, \mathrm{CSS}_i, f_{80,i}) + \delta_t(\mathbf{z}_i) + \varepsilon_{t,i}, \tag{4}$$

$$\delta_t(\mathbf{z}) = d_{t0} + d_{t1} z_{\mathrm{CSS}} + d_{t2} z_{f_{80}}, \tag{5}$$

$$d_{tj} \sim N(0, \tau_t^2), \qquad \varepsilon_{t,i} \sim N(0, \sigma_t^2), \tag{6}$$

with $\mathbf{z}$ the setting and feed f_{80} standardized on the training surveys, $\tau = 140$ t/h and 36 kW, and $\boldsymbol{\theta}$ the eight engine scalars of Table II. Given $\boldsymbol{\theta}$ the discrepancy and the noise are Gaussian, so the discrepancy is integrated out exactly, $\mathbf{y}_t \sim N\big(f_t(\boldsymbol{\theta}),\, \sigma_t^2\mathbf{I} + \tau_t^2\mathbf{Z}\mathbf{Z}^\top\big)$, and the sampler explores ten dimensions (the eight scalars and the two noise scales); the discrepancy coefficients are then drawn from their exact Gaussian conditional. The noise has $\ln \sigma_t \sim N(\ln 60\ \mathrm{t/h}, 1)$ and $N(\ln 15\ \mathrm{kW}, 1)$. The engine is its own simulator, so no emulator is used.

The sampler is the ensemble sampler `emcee` 3.1.6 [16] with differential-evolution moves (`DEMove` 0.8 and `DESnookerMove` 0.2) [13], [14]: 20 walkers, 7000 steps, 1000 discarded, thinned by 10, seeded. The affine-invariant stretch move of Goodman and Weare [15] was tried first and did not converge: its autocorrelation time stayed above 100 steps and grew with the chain, along the parameters the surveys do not identify. A posterior is recorded as converged when the post-burn chain is at least 50 integrated autocorrelation times long. The predictive interval of a held-out survey is the 10th to 90th percentile of 20 predictive draws per retained posterior sample.

V. EVALUATION PROTOCOL

The protocol is shared by every method and is called CrusherCal.

- **Strict leave-one-survey-out.** For each of the ten surveys, every fitted quantity (the capacity base, the power-form constants, the ridge and free-form coefficients, the line, M_{ic} and the no-load power, and the whole CrusherBayes posterior) is refit on the other nine, and the held-out survey enters only the prediction. A test plants an absurd target in one survey and shows that every other fold moves and its own does not.
- **Leaky folds.** The same fits on all ten surveys expose the optimism of an evaluation that lets the held-out survey into the fit.
- **Controls.** A constant-mean predictor, and a label-shuffle control that must not beat the backbone.
- **Nested 80% band.** The band of the calibrated backbone is the prediction $\pm 1.2816\,\sigma$, with σ of each survey taken from the held-out residuals of the other nine surveys only. A band whose σ comes from the same ten residuals it is checked against is in sample, and is reported only to show the difference.
- **Scores.** MAPE in percent, RMSE in the target's unit, R^2, and for the bands the share of surveys inside the interval (coverage) and the mean width in the target's unit. With $n = 10$, one survey moves a coverage by 0.1, and a difference below about 3 MAPE points is within noise.
- **Stopping criterion stated before the run.** CrusherBayes is dropped as a predictor if (a) its power MAPE is not below that of the line by at least 3 points and its throughput MAPE is not below that of the calibrated backbone by at least 3 points, and (b) on either target its coverage lies outside 0.7 to 0.9 or its mean band is wider than the band of the calibrated backbone. The identifiability table is reported in either case.

VI. RESULTS

A. The engine across the circuit cases

The engine reproduces the size-reduction ladder of a crushing circuit (Fig. 1). The gyratory case G01 gives a coarse product ($P_{80} = 153.6$ mm) at high throughput (4629 t/h), the jaw case J01 an intermediate one (126.4 mm, 923 t/h), the secondary cone S01 27.7 mm at 643 t/h, the tertiary cone T01 7.1 mm at 240 t/h and the short-head cone H01 4.6 mm at 216 t/h. This shows that the coupled models behave as the cited models are known to behave; it is not a validation against a plant.

B. Benchmark on the ten HP500 surveys

Table III and Fig. 2 give the benchmark. On throughput the methods that use the setting and the capacity physics sit between 12.00% and 13.67%, inside the noise of $n = 10$, and all beat the constant mean (18.29%); the label-shuffle control (25.04%) is worse than the backbone, as it must be. CrusherBayes has the lowest throughput MAPE (12.00%) and the narrowest band (439 t/h against 474 t/h for the backbone, 7% narrower), at coverage 0.80.

On power the picture is different. The calibrated backbone, whose power is the Bond draw refit in the paper's form, is worse than the constant mean (25.94% against 24.48%) because the Bond draw is nearly flat while the measured power is not. The setting-only line is the best on power (12.78%), followed by the free-form fit (14.03%), CrusherBayes (14.31%) and Morrell with a no-load term (15.21%). CrusherBayes is

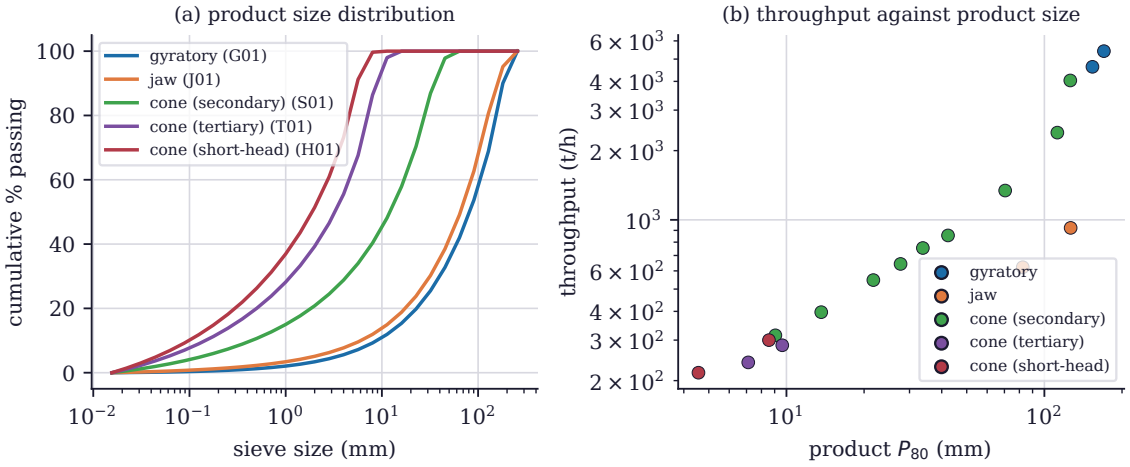


Fig. 1. The coupled engine over the circuit cases. **(a)** Product size distributions of one gyratory (G01), jaw (J01), secondary cone (S01), tertiary cone (T01) and short-head cone (H01) case. **(b)** Product P_{80} against throughput for the 17 circuit cases, coloured by machine type.

TABLE III
STRICT LEAVE-ONE-SURVEY-OUT SCORES OF EVERY METHOD ON THE SAME TEN HP500 SURVEYS. MAPE IN PERCENT, RMSE IN THE UNIT OF THE TARGET, COVERAGE OF THE NESTED 80% BAND AND ITS MEAN WIDTH IN THE UNIT OF THE TARGET. A DASH MEANS THE METHOD GIVES NO BAND OR THE SCORE IS NOT DEFINED. THE LINE, THE FREE-FORM FIT AND THE CALIBRATED BACKBONE CARRY THE NESTED BAND OF THE BACKBONE RESIDUALS; CRUSHERBAYES CARRIES ITS POSTERIOR PREDICTIVE INTERVAL.

	Throughput (t/h)				Power (kW)			
Method	MAPE	RMSE	Cov.	Width	MAPE	RMSE	Cov.	Width
Constant mean	18.29	258.52	–	–	24.48	68.50	–	–
Label-shuffle control	25.04	322.81	–	–	40.19	110.53	–	–
Calibrated backbone	12.09	176.67	0.80	473.76	25.94	78.52	0.90	209.79
Backbone, bounded residual	13.45	189.49	–	–	25.37	92.09	–	–
Free-form fit	12.18	174.42	0.80	468.51	14.03	48.67	0.80	129.82
Setting-only line	13.67	177.75	0.80	478.76	12.78	44.56	0.80	119.60
Morrell	–	–	–	–	38.08	111.67	–	–
Morrell with no-load term	–	–	–	–	15.21	46.53	–	–
CrusherBayes	12.00	168.57	0.80	439.12	14.31	44.68	0.90	117.74

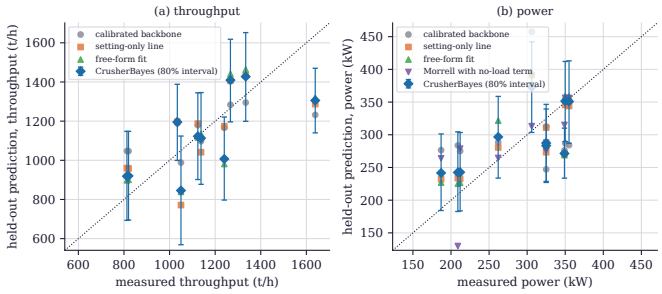


Fig. 2. Held-out predictions against the measured values on the ten surveys. Markers of the calibrated backbone, the free-form fit, the setting-only line and (for power) Morrell with a no-load term, and the CrusherBayes mean with its 80% interval. The dotted line is parity.

1.53 points behind the line, which is within the noise of $n = 10$ and is not a gain. Its power band (118 kW) is 44% narrower than the band of the calibrated backbone (210 kW) at coverage 0.90 and equals the width of the line's band (120 kW), which has coverage 0.80. The stopping criterion of Section V is evaluated as written: condition (a) holds (power -1.53 points against the line, throughput $+0.09$ points against the backbone), and condition (b) does not (coverage 0.80 and 0.90, both inside 0.7 to 0.9, and both bands narrower than the backbone's). The rule does not drop the method. The result is a calibrated band, not a predictive gain over the line.

C. Optimism of an evaluation that leaks

Table IV gives the same methods when the held-out survey is allowed into the fit. The fits that adapt to the data gain most: the bounded residual falls from 13.45% to 8.51% on throughput and from 25.37% to 12.00% on power, and the free-form fit from 14.03% to 9.46% on power. The calibrated backbone, with one scalar, gains less (12.09% to 10.95% on throughput). The label-shuffle control on power falls from 40.19% to 15.50%, which shows how much of the apparent skill of a leaky evaluation is the fit absorbing the survey.

D. Morrell's size-specific energy

Without a no-load term the fit is one positive scalar, M_{ic} between 3.70 and 4.21 kWh/t across the folds, and the held-out error is 38.08%, the largest of the size-aware models. With a no-load term the error falls to 15.21%, but the fit is not Morrell's physics: in every fold the fitted M_{ic} is

TABLE IV
MAPE (%) WITH THE HELD-OUT SURVEY OUTSIDE THE FIT (STRICT) AND INSIDE IT (LEAKY), SAME TEN SURVEYS.

	Throughput		Power	
Method	strict	leaky	strict	leaky
Constant mean	18.29	16.46	24.48	22.03
Label-shuffle control	25.04	21.18	40.19	15.50
Calibrated backbone	12.09	10.95	25.94	19.28
Bounded residual	13.45	8.51	25.37	12.00
Free-form fit	12.18	8.24	14.03	9.46
Setting-only line	13.67	10.60	12.78	10.03
Morrell	–	–	38.08	33.81
Morrell, no-load term	–	–	15.21	11.50

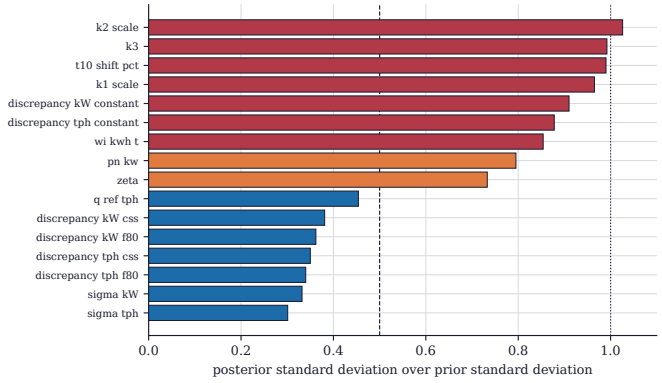


Fig. 3. Posterior over prior standard deviation of each CrusherBayes parameter, fit to all ten surveys. Below 0.5 (dashed line) the surveys identify the parameter; near 1 (dotted line) it stays at its prior.

negative (-6.29 to -4.16 kWh/t) with a no-load power of 591 to 727 kW, about twice the largest measured power. The measured power falls as the throughput rises, and the design column rises with it, so the least-squares line has a negative slope. The variant is a descriptive line on that column and not an estimate of M_{ic}. Both variants are worse than the setting-only line; the difference of 2.4 points for the no-load variant is within the noise, and the 25 points of the variant without it are not. The size-specific energy adds no predictive skill on these ten surveys. An SMC Test of the ore and a no-load reading of the machine would settle the fit.

E. What the ten surveys identify

Fig. 3 gives the contraction of the posterior against the prior. The capacity base Q_T (contraction 0.45, posterior mean 646 t/h against the nominal 647), the two noise scales (0.30 and 0.33) and the discrepancy slopes on the setting and on f_{80} (0.34 to 0.38) are identified. The posterior noise is 145 t/h and 38.8 kW, much larger than the prior medians (60 t/h and 15 kW), and the discrepancy slope on the setting is large ($+100.8$ t/h and -57.1 kW per standardized unit): the engine's dependence on the setting differs from the measured one, in power by about 57 kW per standard deviation of the setting. The work index, ζ and P_n contract weakly (0.85, 0.73 and 0.80; ζ and W_i enter the power only as a product). The K_1 and K_2 multipliers, K_3, the t_{10} shift and the discrepancy constants stay at their prior (contraction 0.88 to 1.03). The

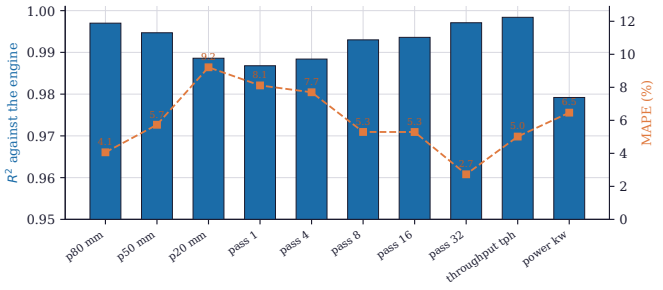


Fig. 4. The surrogate against the engine, per output, on the independent held-out draw (2196 points): R^2 (bars) and MAPE (line), computed through onnxruntime on the shipped float32 graph.

TABLE V
SURROGATE ACCURACY PER OUTPUT AGAINST THE ENGINE ON THE HELD-OUT DRAW.

Output	R^2	MAPE (%)	MAPE int8 (%)
P_{80}	0.9970	4.06	4.57
P_{50}	0.9947	5.73	6.13
P_{20}	0.9886	9.21	9.48
passing 1 mm	0.9868	8.12	8.33
passing 4 mm	0.9884	7.70	7.91
passing 8 mm	0.9930	5.30	5.64
passing 16 mm	0.9936	5.29	5.54
passing 32 mm	0.9971	2.73	2.87
throughput	0.9984	5.02	5.43
power	0.9792	6.47	6.43

scalars that set the product size are not identified because the two measured targets carry the product size only through the Bond term of the power, and the power follows the setting, not the product size. Ten surveys without a measured product size cannot calibrate the population balance.

All eleven posteriors (ten folds and the fit to all surveys) are converged under the criterion of 50 autocorrelation times: the largest autocorrelation time is 48 to 85 steps and the 6000 post-burn steps are 71 to 125 times it, with acceptance fractions between 0.17 and 0.20.

F. Learned models

The surrogate reproduces every output of the engine at R^2 between 0.979 (power) and 0.998 (throughput) and a MAPE of 2.7% to 9.2%, worst on the fine P_{20} tail (Fig. 4, Table V); the product P_{80} is monotone in the setting on all 15 curves tested, and the 8-bit quantized graph stays within about half a percentage point of the float32 one. The autoencoder raises an alarm on 1% of the held-out points (threshold 0.2445) and on 27.96% of planted points moved outside the training range, against 0% and 13.89% for the rule that flags only physically implausible inputs. Neither learned model is held to the HP500 surveys; the comparison is with the engine.

VII. LIMITS AND NEGATIVE RESULTS

- **Sample size.** $n = 10$ surveys, one machine, one ore. A difference below about 3 MAPE points is noise, and one survey moves a coverage by 0.1. The coverage of 0.80 and 0.90 are coarse estimates.

- **CrusherBayes does not beat the setting-only line on power** (14.31% against 12.78%), and beats the calibrated backbone on throughput by 0.09 points. Its contribution on these data is a calibrated and narrower band and the identifiability result.
- **The population-balance scalars are not identified.** K_1, K_2, K_3 and the t_{10} shift stay at their prior, because P_{80} is not measured. Their posterior is the prior and must not be read as an estimate.
- **The priors' widths are the author's.** No published range of these fitted scalars for this ore was found.
- **Bond on the measured power.** The Bond draw does not track the measured power spread; the paper's power constants applied to it give 47.58% in sample.
- **Morrell.** M_{ic} and the no-load power are fitted, not measured; the negative fitted M_{ic} of the no-load variant means that variant is descriptive. The equations are taken from the guideline; the primary paper was not opened. Only the conventional-crusher branch of the method is used.
- **Scope of the engine.** The studio couples calibrated models, not a plant. The synthetic machines are not calibrated. The three-dimensional chamber view is a kinematic animation and not a discrete-element solve. The learned models emulate the engine, not a plant.
- **Discrete-element simulation.** The package that is planned for it is not validated and no result of it is reported.
- **Prior art.** Searches found no Bayesian calibration with structural discrepancy of a cone-crusher population-balance model evaluated held out; they were not exhaustive (Scopus and Web of Science were not searched).

VIII. Conclusion

On ten industrial surveys of one cone crusher, a one-feature line on the closed-side setting is a hard baseline: it reaches 13.67% on throughput and 12.78% on power held out, and the calibrated physics, a free-form fit, Morrell's size-specific energy and a Bayesian calibration with structural discrepancy all land within the noise of it or worse. CrusherBayes gives a posterior predictive band of the right coverage that is 44% narrower than the band of the calibrated engine on power and 7% narrower on throughput, shows that ten surveys identify the capacity scale, the noise and the setting dependence of the discrepancy, and shows that they cannot identify the classification and breakage scalars. Measuring the product size distribution of each survey is the data that would change that.

Appendix A
Reproduction

Commands: From the repository root:

- `python data-pipeline/run.py hp500` bakes the backbone, the leave-one-survey-out document and the CrusherBayes document (the sampler runs in this step only);
- `python data-pipeline/run.py models` trains and scores the learned models;
- `python manuscripts/comminution-studio/figures/make_figs.py` regenerates every figure from the committed artifacts;
- `latexmk -pdf main.tex`, run in `manuscripts/comminution-studio/tex`, builds the report.

Versions: Python 3.13 (3.12 in continuous integration), numpy 2.4.6, emcee 3.1.6, torch 2.12.1, onnx 1.22.0, onnxruntime 1.27.0, matplotlib 3.11.2, Node 24. The artifacts record the engine build 0.11.000 and the seed 42.

Artifact hashes (SHA-256):

```
hp500-surveys.csv    ab9609a7...39b19f7a
backbone.json        204ebb46...9ce94fc9
loo.json             269b3148...ae98edba
crusherbayes.json    2ad1ae73...7e0fb96c
model-metrics.json   1f77b2d4...e1e6f11f
```

The full digests are in `manuscripts/comminution-studio/data/sha256.txt`; the sampler input digest recorded in `crusherbayes.json` is `483d24c5...b0ce605a`.

Notebooks: `notebooks/en` and `notebooks/es` hold the same notebooks: getting started, size distributions, breakage and classification, capacity and power, the real surveys, the held-out validation, the learned models and CrusherBayes.

References

[1] Cochilco, *Consumo de energía en la minería del cobre, actualización año 2024*. Santiago: Comisión Chilena del Cobre, 2025. https://www.cochilco.cl/web/download/991/2025/15151/consumo-de-energia-en-la-mineria-del-cobre-actualizacion-ano-2024.pdf

[2] Cochilco, *Proyección de consumo de energía eléctrica de la minería del cobre, período 2025–2034* (DEPP 03/2026). Santiago: Comisión Chilena del Cobre, 2026. https://www.cochilco.cl/web/download/1014/2026/15677/proyeccion-de-consumo-de-energia-electrica-de-la-mineria-del-cobre-periodo-2025-20.pdf

[3] B. K. N. Rocha, T. M. Campos, J. Silva, and L. M. Tavares, "Fit-for-purpose model of HP500 cone crusher in size reduction of itabirite iron ore," *Minerals*, vol. 14, no. 9, art. 919, 2024. doi:10.3390/min14090919.

[4] W. J. Whiten, "The simulation of crushing plants with models developed using multiple spline regression," *Journal of the South African Institute of Mining and Metallurgy*, vol. 72, pp. 257–264, 1972.

[5] T. J. Napier-Munn, S. Morrell, R. D. Morrison, and T. Kojovic, *Mineral Comminution Circuits: Their Operation and Optimisation*. Brisbane: Julius Kruttschnitt Mineral Research Centre, 1996.

[6] C. M. Evertsson, "Modelling of flow in cone crushers," *Minerals Engineering*, vol. 12, no. 12, pp. 1479–1499, 1999. doi:10.1016/S0892-6875(99)00136-3.

[7] C. M. Evertsson, *Cone Crusher Performance*, Ph.D. dissertation, Chalmers University of Technology, Göteborg, Sweden, 2000.

[8] F. C. Bond, "The third theory of comminution," *Transactions of the American Institute of Mining, Metallurgical and Petroleum Engineers*, vol. 193, pp. 484–494, 1952.

[9] S. Morrell, "Predicting the overall specific energy requirement of crushing, high pressure grinding roll and tumbling mill circuits," *Minerals Engineering*, vol. 22, no. 6, pp. 544–549, 2009. doi:10.1016/j.mineng.2009.01.005.

[10] Global Mining Standards and Guidelines Group, *Morrell Method for Determining Comminution Circuit Specific Energy and Assessing Energy Utilization Efficiency of Existing Circuits*, Annex B. https://www.smctesting.com/documents/Morrell_method_for_determining_comminution_circuit_specific_energy_and_assessing_energy_utilization_efficiency_of_existing_circuits.pdf

[11] E. J. Y. Koh, E. Amini, G. J. McLachlan, and N. Beaton, "Utilising a deep neural network as a surrogate model to approximate phenomenological models of a comminution circuit," *Minerals Engineering*, vol. 170, art. 107026, 2021. doi:10.1016/j.mineng.2021.107026.

[12] M. C. Kennedy and A. O'Hagan, "Bayesian calibration of computer models," *Journal of the Royal Statistical Society: Series B*, vol. 63, no. 3, pp. 425–464, 2001. doi:10.1111/1467-9868.00294.

[13] C. J. F. ter Braak, "A Markov chain Monte Carlo version of the genetic algorithm Differential Evolution: easy Bayesian computing for real parameter spaces," *Statistics and Computing*, vol. 16, pp. 239–249, 2006. doi:10.1007/s11222-006-8769-1.

[14] C. J. F. ter Braak and J. A. Vrugt, "Differential Evolution Markov Chain with snooker updater and fewer chains," *Statistics and Computing*, vol. 18, pp. 435–446, 2008. doi:10.1007/s11222-008-9104-9.

[15] J. Goodman and J. Weare, "Ensemble samplers with affine invariance," *Communications in Applied Mathematics and Computational Science*, vol. 5, no. 1, pp. 65–80, 2010. doi:10.2140/camcos.2010.5.65.

[16] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman, "emcee: The MCMC hammer," *Publications of the Astronomical Society of the Pacific*, vol. 125, pp. 306–312, 2013. doi:10.1086/670067.